Exploratory Analysis of MMP11 and SCARA5 Counts in Breast Cancer

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1 Introduction

This project explores RNA-seq gene counts and sample metadata from The Cancer Genome Atlas Breast Invasive Carcinoma (TCGA-BRCA) cohort, whose data are available through the National Cancer Institute's Genomic Data Commons [1]. I used the course-provided count matrix and metadata files. The matrix contains 60,660 genes across 1,231 samples: 1,111 primary tumors, 113 normal tissue samples, and seven metastatic samples.

To guide gene selection, I first explored genes with large observed differences between normal and primary tumor samples. An exploratory heatmap helped me examine these patterns across individual samples. I selected *MMP11* as my main gene because it generally showed higher expression in primary tumors, and *SCARA5* as a comparison gene because it showed the opposite pattern.

MMP11 encodes matrix metallopeptidase 11, also called stromelysin-3. It was originally identified in stromal cells surrounding invasive breast carcinomas, making it relevant to the study of the tumor microenvironment [2]. *SCARA5* encodes scavenger receptor class A member 5. Previous research reported reduced SCARA5 expression in breast tumors compared with matched noncancerous tissue [3].

My objective was to describe the distributions of these genes' counts and compare patterns across tissue types. Following peer feedback, I highlighted the metastatic samples to examine their positions relative to normal and primary tumor samples. I also added PAM50 molecular subtype annotations to the heatmap to provide further context for variation among primary tumors.

Because gene selection followed exploration of this same dataset, the findings are descriptive and exploratory.

2 Methods

2.1 Data preparation

I imported the course-provided count matrix and sample metadata into R. Rows in the count matrix represent genes, and columns represent samples. I matched the sample barcodes in the metadata to the corresponding count columns so that each sample's gene counts were associated with the correct tissue type.

The analysis included 113 normal tissue samples, 1,111 primary tumor samples, and seven metastatic samples. I extracted counts for *MMP11* (ENSG00000099953.10) and *SCARA5* (ENSG00000168079.17) for the two-gene comparisons.

2.2 Exploratory gene selection

For the initial comparison of normal and primary tumor samples, I converted raw counts to counts per million (CPM) by dividing each count by its sample's total count across all genes and multiplying by one million. I retained genes with at least one CPM in at least 10 samples and transformed their values using $\log_2(\text{CPM} + 1)$.

I calculated the difference between the median transformed values in primary tumor and normal tissue samples for each retained gene. I ranked genes by the absolute size of this difference and explored the top 20 candidates using a heatmap. This exploration guided my selection of *MMP11* and *SCARA5*.

2.3 Summary statistics and visualizations

For each selected gene, I calculated the sample size, mean, median, standard deviation, minimum, and maximum of the raw counts across all samples.

I used a histogram to examine the distribution of *MMP11* counts within each tissue group, a scatter plot to examine the relationship between the two genes, and box plots to compare their distributions across tissue types. These plots used $\log_{10}(\text{raw count} + 1)$ to make the wide range of counts easier to display. Adding one allowed samples with zero counts to remain in the plots. This transformation does not normalize counts for differences in sequencing depth.

Following peer feedback, I highlighted the seven metastatic samples as orange triangles in the scatter plot. I also displayed individual metastatic samples on the box plots because this group was small.

As an additional plot type, I created empirical cumulative distribution function (ECDF) plots using `stat_ecdf()` in ggplot2 [4]. At each count value, an ECDF shows the proportion of samples with values at or below that value. These plots allowed me to compare distributions across tissue groups without choosing histogram bins.

2.4 Heatmap

The final heatmap included 10 unique genes: *MMP11*, *SCARA5*, and eight additional genes from the exploratory candidate list. For these genes, I calculated CPM using sample totals from the complete count matrix, applied $\log_2(\text{CPM} + 1)$, and standardized each gene across samples using a row z-score.

For each gene, the z-score was calculated by subtracting its mean transformed value across samples and dividing by its standard deviation. Positive values therefore indicate values above that gene's mean, while negative values indicate values below its mean.

I grouped samples by tissue type and used hierarchical clustering to organize genes and samples within each tissue group. Annotation bars identified tissue type and the PAM50 molecular subtype

recorded in the metadata. Missing subtype information was displayed as "Not available."

2.5 Software and scope

I performed the analysis using R version 4.6.1. Data preparation used tidyverse 2.0.0, including dplyr 1.2.1, tidyr 1.3.2, and tibble 3.3.1. Plots used ggplot2 4.0.3 and scales 1.4.0. The heatmap used ComplexHeatmap 2.28.0 and circlize 0.4.18. The accompanying R Markdown document used knitr 1.51 and rmarkdown 2.32.

The analysis was descriptive and exploratory. I did not perform hypothesis tests or a formal differential expression analysis. Samples were compared by tissue group rather than as matched patient pairs.

3 Results

3.1 Summary statistics

Table 1 summarizes the raw counts for both genes across all 1,231 samples.

Table 1: Summary statistics for raw RNA-seq counts across all samples.

Gene	Samples	Mean	Median	SD	Minimum	Maximum
MMP11	1231	24676.13	14465	29551.41	51	281893
SCARA5	1231	792.20	70	2736.32	0	30188

Across all samples, *MMP11* had a mean count of 24,676.13 and a median of 14,465, while *SCARA5* had a mean of 792.20 and a median of 70. For both genes, the mean exceeded the median, consistent with right-skewed distributions in which a relatively small number of high counts increased the mean.

Counts ranged from 51 to 281,893 for *MMP11* and from zero to 30,188 for *SCARA5*. These pooled statistics describe variation across the complete dataset; the following plots examine how counts differ between tissue groups.

3.2 Distribution of MMP11 counts

Figure 1 shows the distribution of log-transformed *MMP11* raw counts separately for each tissue group.

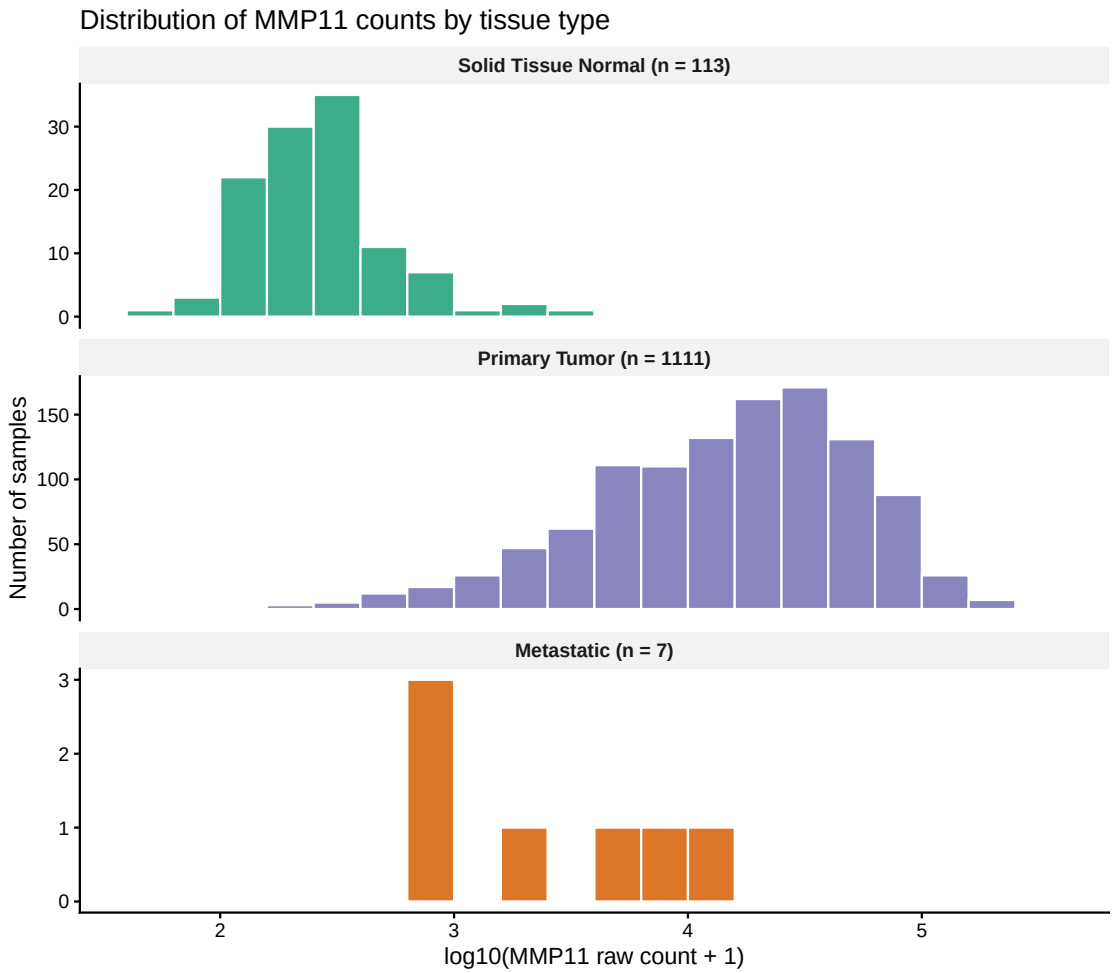


Figure 1: Distribution of log-transformed *MMP11* raw counts in normal tissue, primary tumor, and metastatic samples. Values are displayed as $\log_{10}(\text{raw count} + 1)$. Panels share the same horizontal scale but use different vertical scales because the groups have different sample sizes.

The primary tumor distribution was shifted toward higher counts compared with normal tissue, although the groups overlapped. Primary tumors also showed a wider spread of log-transformed counts, indicating variation among tumor samples.

The seven metastatic samples were spread across an intermediate range relative to the normal and primary tumor distributions. With so few metastatic samples, the histogram provides limited information about the shape of this group's distribution.

Because the panels use different vertical scales, bar heights should be interpreted within each panel. The shared horizontal scale allows comparison of the locations and ranges of counts across tissue groups.

3.3 Relationship between MMP11 and SCARA5 counts

Figure 2 displays both genes' log-transformed raw counts for each sample. Following peer feedback, I highlighted the seven metastatic samples as orange triangles to make their positions easier to identify.

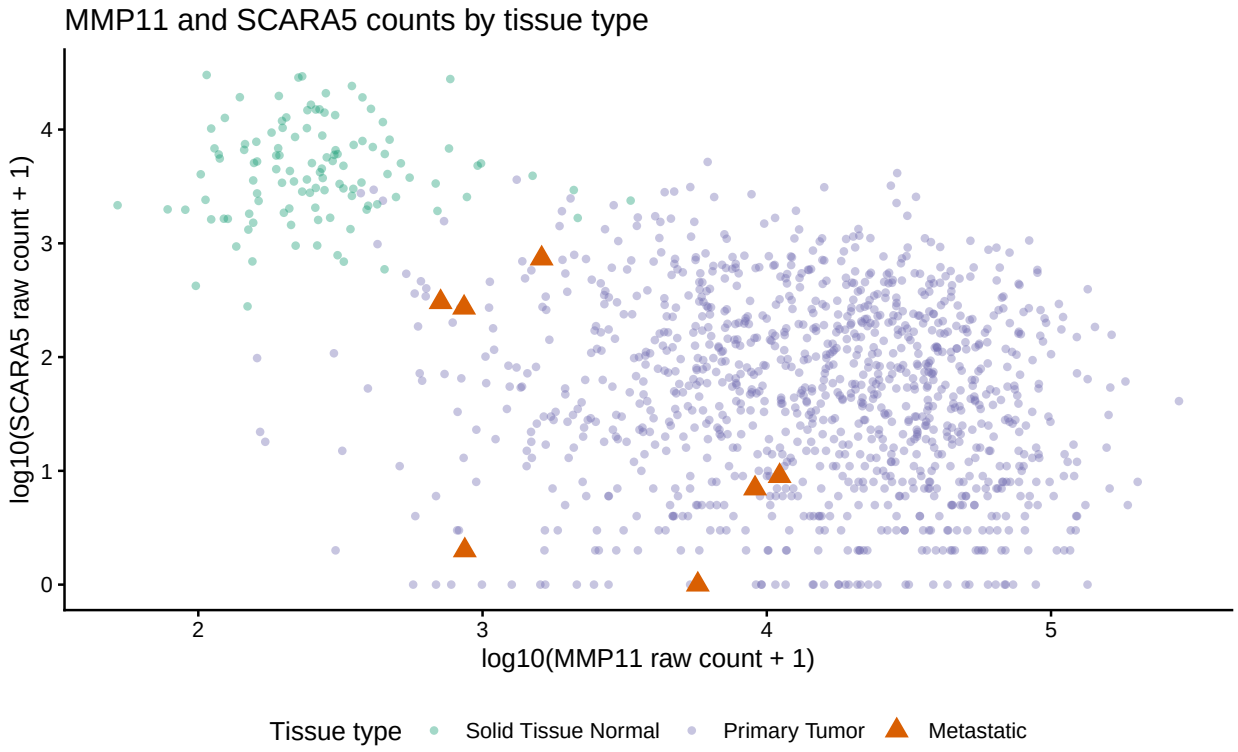


Figure 2: Relationship between log-transformed *MMP11* and *SCARA5* raw counts. Both axes use $\log_{10}(\text{raw count} + 1)$. Each point represents one sample. Normal tissue samples are green, primary tumors are purple, and metastatic samples are highlighted as orange triangles.

Normal tissue samples generally appeared toward the upper left, with lower *MMP11* counts and higher *SCARA5* counts. Primary tumors generally appeared farther to the right and lower on the plot, showing the opposite pattern. The groups overlapped, and primary tumors showed considerable variation.

The metastatic samples overlapped the broad range occupied by primary tumors rather than forming a clearly separate cluster. Their positions varied, with some showing lower *MMP11* counts than most primary tumors. With only seven samples, this observation remains descriptive.

The overall pattern suggests an inverse relationship between the two genes across the combined dataset. However, much of this pattern reflects differences between tissue groups and does not establish that one gene regulates the other.

3.4 Comparison of counts across tissue types

Figure 3 summarizes the log-transformed raw counts for both genes by tissue type.

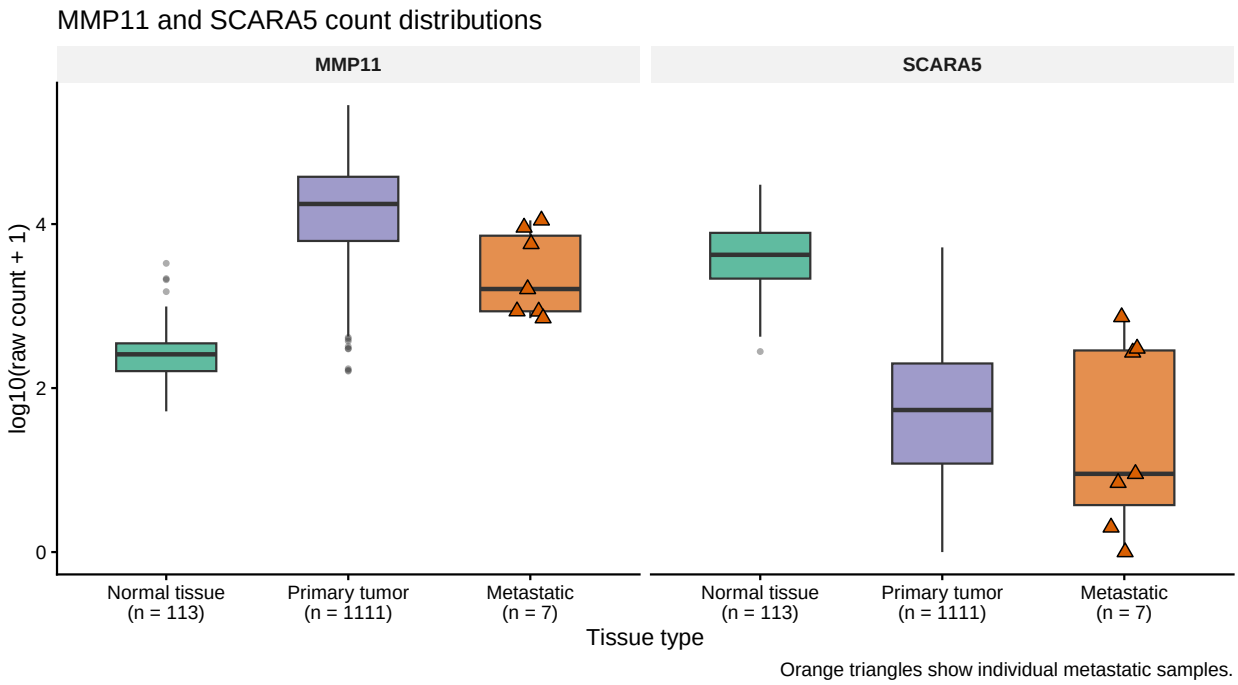


Figure 3: Box plots of $\log_{10}(\text{raw count} + 1)$ for *MMP11* and *SCARA5* across tissue groups. The central line represents the median, and each box contains the middle 50% of values. Whiskers extend to the most extreme observations within 1.5 times the interquartile range of the box. Points beyond the whiskers are shown individually. Orange triangles show the seven individual metastatic samples.

For *MMP11*, primary tumors had a higher median count than normal tissue. The middle 50% of values in these two groups were clearly separated, although some individual samples overlapped. The metastatic median fell between the normal and primary tumor medians.

For *SCARA5*, normal tissue had a higher median count than either tumor group. Primary tumors showed substantial variation, including samples with zero counts. The metastatic median was lower than both other groups, but the seven samples were widely spread.

These plots reinforce the contrasting patterns observed for the two genes. The individual metastatic points also show how much variation can be hidden by a summary box when the group contains very few samples. These comparisons describe the observed data and do not establish statistical significance.

3.5 Patterns across selected genes

Figure 4 shows the expression patterns of 10 selected genes across all samples, including *MMP11* and *SCARA5*.

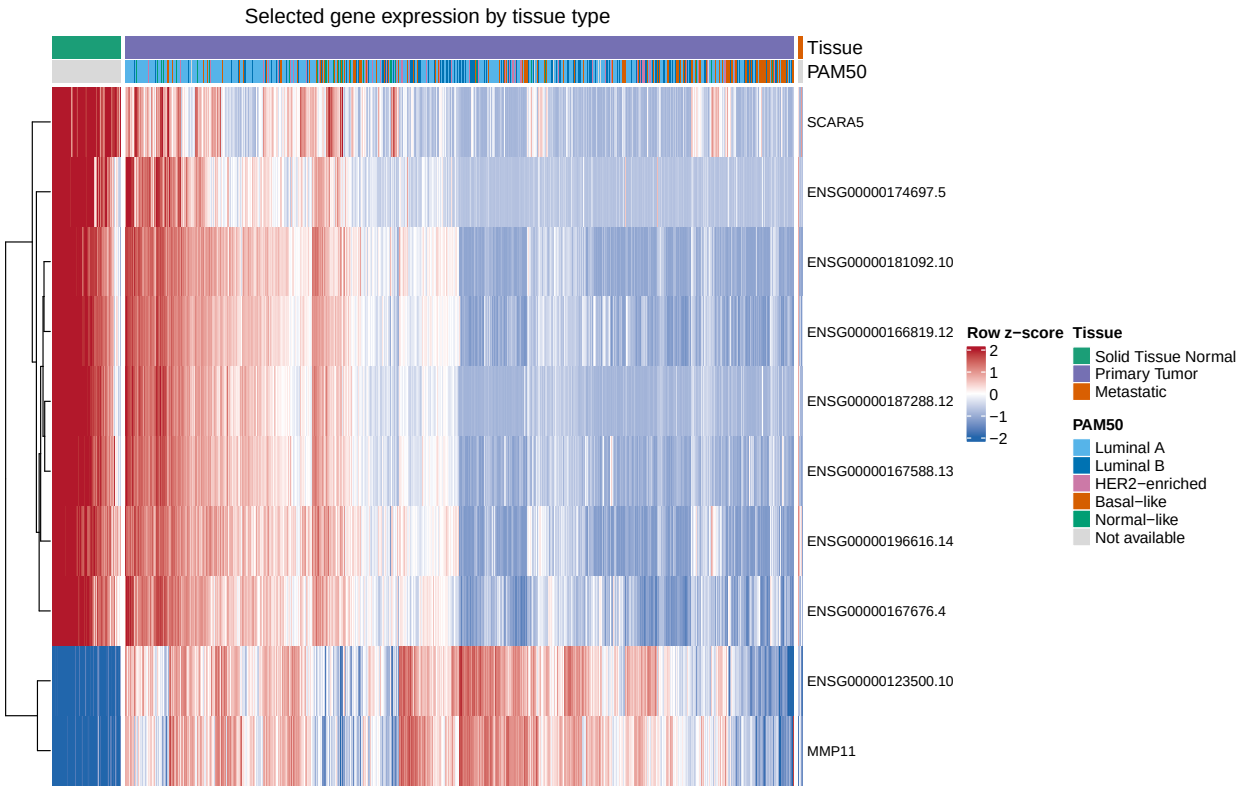


Figure 4: Heatmap of 10 selected genes across 1,231 samples. Values were converted to CPM, transformed using $\log_2(\text{CPM} + 1)$, and standardized within each gene. Red indicates values above that gene's mean across samples, while blue indicates values below its mean. Samples are grouped by tissue type and clustered within each group. Annotation bars identify tissue type and PAM50 molecular subtype; gray indicates unavailable subtype information.

Normal tissue samples generally showed higher standardized values for *SCARA5* and several other selected genes, while *MMP11* showed lower values. Many primary tumors showed the opposite pattern, although there was considerable variation within this group.

The PAM50 annotation provided context for variation among primary tumors. Subtypes were interspersed across the heatmap rather than forming completely separate blocks. The PAM50 category "Normal-like" describes a molecular subtype and should not be confused with the normal tissue sample group.

The metastatic samples form a narrow group at the right edge because there were only seven. Their small number limits interpretation of a shared metastatic expression pattern.

Row standardization highlights relative variation within each gene. Therefore, a red cell for one gene does not necessarily represent a higher absolute expression level than a blue cell for another gene.

Because most genes were selected for large observed differences between normal and primary tumor samples, the contrasts in this heatmap are expected and do not provide independent validation of the gene-selection results.

3.6 Cumulative distributions of gene counts

Figure 5 compares the cumulative distributions of both genes across tissue groups. At any horizontal-axis value, the height of a curve shows the percentage of samples with values at or below that value.

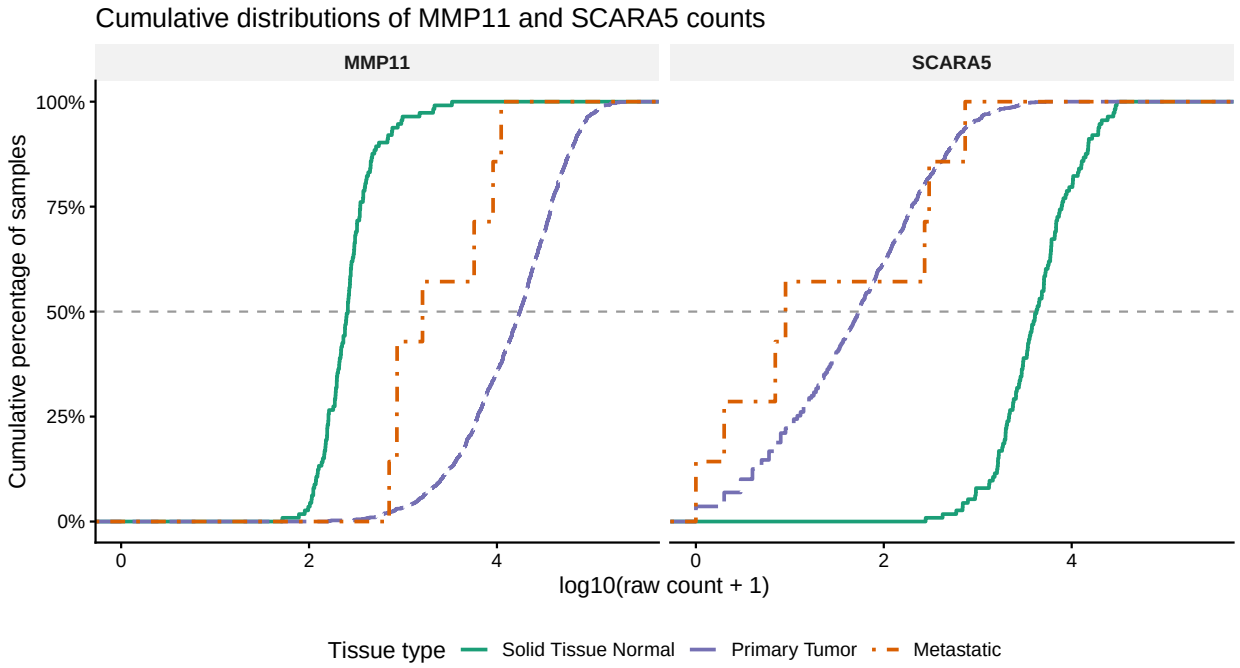


Figure 5: Empirical cumulative distribution functions of $\log_{10}(\text{raw count} + 1)$ for *MMP11* and *SCARA5*, grouped by tissue type. The dashed horizontal line marks 50%, allowing comparison of the group medians. Each metastatic sample contributes approximately 14.3 percentage points to its group's curve, producing larger steps than in the other groups.

For *MMP11*, the primary tumor curve was shifted to the right of the normal tissue curve, indicating generally higher counts in primary tumors. The metastatic curve lay largely between them, consistent with its intermediate median in the box plot.

For *SCARA5*, the normal tissue curve was shifted to the right of both tumor groups, indicating generally higher counts in normal tissue. The metastatic and primary tumor curves crossed, showing that their distributions could not be described by a uniform shift in one direction.

Unlike a histogram, an ECDF does not require a choice of bins. It also expresses each group's distribution as a percentage, making groups with different sample sizes easier to compare. However, the metastatic curve remains based on only seven observations and should be interpreted cautiously.

4 Conclusion

This exploratory analysis identified contrasting patterns for *MMP11* and *SCARA5* across tissue groups. Primary tumors generally had higher *MMP11* counts and lower *SCARA5* counts than normal tissue. These patterns were visible across several complementary visualizations, although individual samples overlapped and tumor samples showed considerable variation.

In response to peer feedback, I highlighted the seven metastatic samples in the scatter plot. They overlapped the broad range occupied by primary tumors rather than forming a clearly separate cluster. Their small number limits conclusions about metastatic tissue more broadly.

Several limitations affect interpretation. The tissue groups differed greatly in sample size, and some samples came from the same patients. The histogram, scatter plot, box plots, and ECDFs used log-transformed raw counts, which do not adjust for differences in sequencing depth. The heatmap used CPM, which accounts for sample count totals but does not address all potential technical or biological differences.

Finally, gene selection and visualization used the same dataset. These findings therefore describe observed patterns rather than providing independent validation, evidence of causation, or a validated diagnostic test. Further work could use formal RNA-seq normalization, account for patient relationships and relevant covariates, and evaluate the selected genes in an independent dataset.

References

[1] National Cancer Institute. *TCGA-BRCA: Breast Invasive Carcinoma*. Genomic Data Commons Data Portal. https://portal.gdc.cancer.gov/projects/TCGA-BRCA. Accessed October 18, 2026.

[2] Basset, P., Bellocq, J. P., Wolf, C., et al. (1990). A novel metalloproteinase gene specifically expressed in stromal cells of breast carcinomas. *Nature*, 348, 699–704. https://doi.org/10.1038/348699a0.

[3] Ulker, D., Ersoy, Y. E., Gucin, Z., Muslumanoglu, M., and Buyru, N. (2018). Downregulation of SCARA5 may contribute to breast cancer via promoter hypermethylation. *Gene*, 673, 102–106. https://doi.org/10.1016/j.gene.2018.06.036.

[4] ggplot2. *Compute empirical cumulative distribution: stat_ecdf*. Package documentation. https://ggplot2.tidyverse.org/reference/stat_ecdf.html. Accessed October 18, 2026.